dark energy in the form of a cosmological constant, in the appropriate units, the pressure is exactly the negative of the density.

We usually talk about the relationship between a substance's density and its pressure using a number called the *equation of state* parameter, written as *w*—it's equal to pressure divided by energy density, in units in which that comparison makes sense. Here, we're interested in the equation of state of dark energy, which, given enough time, will be the equation of state of the whole universe, since dark energy becomes more and more important in the expanding universe as everything else dilutes away. If the measured value of w = -1 exactly, that tells you that the pressure and the density are exactly opposite, and dark energy is a cosmological constant. Since the energy density in a cosmological constant is always positive, at first glance it seems as though it should act just like matter and amp up the gravity that slows down the expansion of the universe. But because the negative pressure is given a heavier weight in the equations, all a cosmological constant ends up doing is contributing toward accelerating cosmic expansion.

At least it does so in a predictable way. A cosmological constant, with w = -1, has a total energy density that is exactly constant over time as the universe expands, without increasing or decreasing. For dark energy with any other value of w, this is no longer the case. So it's important to figure out what we're really dealing with here.

In the years after dark energy was first discovered, it was clear that *something* was making the expansion of the universe accelerate, which meant there had to be something out there with negative pressure. It turns out that anything that has a value of w less than -1/3 gives you both negative pressure and accelerated expansion. But knowing the value of w could tell us whether dark energy is a true cosmological constant (w = -1 always), or some kind of dynamical dark energy whose influence on the universe might change over time. So astronomers went looking for a way to determine the value of w exactly. If dark energy turned out not to be a cosmological constant, this would indicate that we had not only discovered a new kind of physics acting on the universe, but one with the added bonus of being something even Einstein hadn't foreseen.[fn1]

For a few years, this was the name of the game: measure w, find out what's going on with dark energy. Measurements were made, papers were